HU Collection
ESAT Mock Set F: Chemistry

Worked Solutions

Kevin Hu 29 October 2026

Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	C	D	B	F	D	B	B	C	F	E

Q	11	12	13	14	15	16	17	18	19	20
Ans	A	E	B	E	A	D	B	A	C	C

Q	21	22	23	24	25	26	27			
Ans	F	E	F	D	C	A	A			

Each solution gives an intended route and names the trap the question was built around. Diagrams show the configuration each solution works with.

Question 1. Answer **C**.

Compare equal numbers of molecules. One cubic centimetre of liquid has mass 0.84 g, which is $0.84/28 = 0.030$ mol of N_2. As a gas at room temperature and pressure this occupies $0.030 \times 24\,000 = 720$ cm^3. So the volume available to each molecule is 720 times larger in the gas.

The key step is a scaling argument. If the molecules are spread evenly, each one sits in its own small cube of space whose side is the distance to its neighbours. Volume scales as the cube of that length, so the spacing grows by $\sqrt[3]{720}$. Since $9^3 = 729$, the factor is about 9. Nothing about the size of a molecule or the Avogadro constant is needed.

The answer 720 compares volumes, not distances. Taking a square root instead of a cube root gives 27. Using 14 instead of 28 for the molar mass (forgetting that nitrogen is diatomic) doubles the volume ratio to 1440; taking no root, a cube root or a square root of that gives 1440, 11 or 38.

Question 2. Answer **D**.

Read the axes rather than the shapes. On a graph of rate against volume collected, each curve starts at the initial rate, which is set by the initial concentration of H^+ ions, and reaches zero rate when the acid is used up. So the point where a curve meets the V axis is the *total* volume of hydrogen, which depends only on the amount of acid, not on its volume or its strength.

All four solutions contain 0.0400 mol of a monoprotic acid, so every one of them gives 0.0200 mol of H_2, which is 480 cm^3: all four curves must end at 480 cm^3, like P.

X has half the concentration of P, so it starts lower and ends at 480: curve S. Z has double the concentration, so it starts higher and ends at 480: curve Q. Ethanoic acid is a weak acid, so at any moment only a small fraction of it is ionised and the rate is far lower than for hydrochloric acid of the same concentration. However, as H^+ ions are used up, more ethanoic acid ionises, so all 0.0400 mol eventually reacts: curve U, which also ends at 480 cm^3.

Curves T and R come from confusing the volume of acid with the amount of acid (more solution, more gas). Curve W comes from thinking that a weak acid gives less hydrogen, and choosing P for Y confuses strength with concentration.

Question 3. Answer **B**.

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A reaction is redox when some element changes its oxidation state, so track every element rather than looking for oxygen being gained.

I: chlorine goes from 0 in Cl_2 to -1 in HCl and $+1$ in HClO. One element is both oxidised and reduced (disproportionation), so I is redox.

II: nitrogen is -3 in NH_3 and still -3 in NH_4^+; it is $+5$ in HNO_3 and still $+5$ in NO_3^-. This is an acid and a base forming a salt, not redox.

III: nitrogen goes from 0 to $+2$ and oxygen from 0 to -2, so III is redox.

IV: nitrogen goes from $+4$ in NO_2 to $+5$ in HNO_3 and to $+2$ in NO, another disproportionation, so IV is redox.

V: sulfur is $+4$ in SO_2 and $+4$ in SO_3^{2-}, and carbon is $+4$ throughout. Although sulfur "gains oxygen", no oxidation state changes: an acidic oxide is simply displacing CO_2 from a carbonate. (In a real scrubber the sulfite is then oxidised by air to sulfate, and that later step is redox.)

VI: oxygen is 0 in both O_3 and O_2. Changing the molecule is not changing the oxidation state, so VI is not redox.

The redox reactions are I, III and IV only. Including V comes from using the oxygen-gain definition without checking oxidation states; including VI confuses a new molecule with a new oxidation state; leaving out I or IV misses the fact that disproportionation is redox.

Question 4. Answer **F**.

The Roman numeral (II) is the oxidation state of the iron inside the complex ion, not the charge of the ion. The ion's charge is $(+2) + 6 \times (-1) = -4$, so it is $[Fe(CN)_6]^{4-}$.

Balancing charges, four Fe^{3+} ions $(+12)$ go with three $[Fe(CN)_6]^{4-}$ ions (-12), giving $Fe_4[Fe(CN)_6]_3$. This contains $4 + 3 = 7$ iron atoms and $3 \times 6 = 18$ nitrogen atoms, so the ratio is $7 : 18$.

Reading (II) as a charge of $2-$ gives $Fe_2[Fe(CN)_6]_3$ and $5 : 18$. Swapping the numbers of the two ions gives $Fe_3[Fe(CN)_6]_4$ and $7 : 24$. Counting only the Fe^{3+} ions gives $4 : 18 = 2 : 9$. Swapping the roles of the two irons, iron(II) ions with an iron(III) complex of charge $3-$, gives $Fe_3[Fe(CN)_6]_2$ and $5 : 12$, and taking the complex ion's charge as $3-$ while keeping Fe^{3+} gives $Fe[Fe(CN)_6]$ and $1 : 3$.

Question 5. Answer **D**.

Assign oxidation states before deciding anything.

In stage 1, titanium is $+4$ in TiO_2 and still $+4$ in $TiCl_4$, so I is false. Stage 1 is a redox reaction, but titanium only swaps its oxide ions for chloride ions; the oxygen leaves as CO. The elements that change are carbon, from 0 to $+2$ (oxidised), and chlorine, from 0 to -1 (reduced). Carbon gives electrons to chlorine, so carbon is the reducing agent and chlorine the oxidising agent: II is true.

In stage 2, titanium goes from $+4$ to 0 and magnesium from 0 to $+2$. The substance that accepts the electrons, $TiCl_4$, is the oxidising agent: III is true. (Being reduced and being the oxidising agent are the same fact seen from two sides.)

Per mole of titanium, 2 mol of magnesium reacts: $2 \times 24 = 48$ g of Mg for 48 g of Ti, so IV is true.

In electrolysis, Mg^{2+} ions gain electrons at the negative electrode (the cathode), so V is false.

The answer is II, III and IV only. The popular slip is to pick I, reading stage 1 as the reduction because the oxide disappears.

Question 6. Answer **B**.

Count hydrogen ions, not formula units. The acid supplied $2 \times 0.0250 = 0.0500$ mol of H^+, because sulfuric acid is diprotic. The back-titration used 0.0200 mol of OH^-, so 0.0300 mol of H^+ was taken up by ammonia, one H^+ per NH_3. Hence 0.0300 mol of NH_3, and so 0.0300 mol of NH_4NO_3.

Each NH_4NO_3 contains *two* nitrogen atoms, but only the ammonium one was released as ammonia. Total nitrogen = 0.0600 mol = 0.840 g, and $0.840/3.00 = 28.0\%$. (As a check, the ammonium nitrate is 2.40 g, leaving 0.60 g of potassium chloride.)

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14.0% counts only the ammonium nitrogen; 80.0% is the percentage of ammonium nitrate. Treating sulfuric acid as monoprotic gives 0.0050 mol of ammonia and 4.67%; taking NaOH and H_2SO_4 to react 1 : 1 in the back-titration gives 0.0100 mol of ammonia and 9.33%; ignoring the back-titration altogether gives 46.7%.

Question 7. Answer **B**.

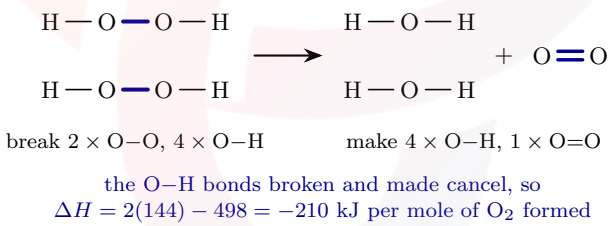
Every bond has two ends, so the number of bonds is half the total number of bond ends. Count bond ends per gram, atom by atom. A hydrogen atom carries one bond end for a mass of only 1, while a carbon atom carries four bond ends for a mass of 12, which is just $\frac{1}{3}$ of a bond end per unit mass. So the substance richest in hydrogen by mass wins, and that is methane.

In moles of bonds per gram: methane has 4 bonds per 16 g, which is 0.25. Poly(ethene) has 3 bonds per CH_2 unit (2 C−H plus one C−C shared between neighbouring units), so $3/14 \approx 0.21$. Diamond has 4 bond ends per C, so 2 bonds per 12 g, which is $\frac{1}{6} \approx 0.17$. Cubic boron nitride has 4 bonds per BN unit of mass 25, which is 0.16. Silicon dioxide has 4 Si−O bonds per 60 g, about 0.067, and white phosphorus has 6 bonds per 124 g, about 0.048.

The trap is to count four bonds for every carbon atom in diamond, which counts each bond twice and gives $\frac{1}{3}$, apparently beating methane. Double counting the C−C bonds in poly(ethene) in the same way gives $4/14 \approx 0.29$.

Question 8. Answer **C**.

The O−H bond energy is not given, and it is not needed. Work per mole of O_2 formed, using the equation as written: two H−O−O−H molecules lose two O−O bonds and four O−H bonds; the products gain four O−H bonds (two water molecules) and one O=O bond. The O−H terms cancel exactly.



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$\Delta H = 2(144) - 498 = -210$ kJ per mole of O_2 formed

$\Delta H = 2(144) - 498 = -210$ kJ per mole of O_2. Energy given to the solution: $100 \times 4.2 \times 5.0 = 2100$ J $= 2.1$ kJ, so $2.1/210 = 0.0100$ mol of O_2, which is $0.0100 \times 24\,000 = 240$ cm^3. The catalyst changes the rate only, not ΔH.

Using 210 kJ per mole of H_2O_2 gives 120 cm^3; using 105 kJ per mole of O_2 gives 480 cm^3. Breaking only one O−O bond in the equation gives $498 - 144 = 354$ kJ per mole of O_2 and about 142 cm^3 (or 71 cm^3 if this is also taken per mole of H_2O_2). Treating 2100 J as 2100 kJ gives 240 dm^3.

Question 9. Answer **F**.

Carbon: $2.64/44 = 0.060$ mol of C atoms. Hydrogen gas: $720/24\,000 = 0.030$ mol, and each OH group gives half a molecule of H_2 with sodium, so there are 0.060 mol of OH groups. So X has exactly one OH group on every carbon atom.

A saturated non-cyclic compound with n carbon atoms and n OH groups is $C_nH_{2n+2}O_n$, with $M_r = 12n+(2n+2)+16n = 30n+2$. The sample contains $0.060/n$ mol of X, so $M_r = 1.86 \div (0.060/n) = 31n$. Then $31n = 30n + 2$ gives $n = 2$: X is ethane-1,2-diol, $HOCH_2CH_2OH$ ($M_r = 62$), and the sample is 0.030 mol.

$2C_2H_6O_2 + 5O_2 \rightarrow 4CO_2 + 6H_2O$, so 0.030 mol of X needs 0.075 mol of O_2, which is $0.075 \times 24 = 1.80$ dm^3. A check by conservation of mass: the products are 2.64 g of CO_2 and $0.090 \times 18 = 1.62$ g of water, so the oxygen used is $2.64 + 1.62 - 1.86 = 2.40$ g $= 0.075$ mol.

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Methanol and propane-1,2,3-triol also have one OH per carbon, but their M_r values (32 and 92) are not $31n$; assuming them gives 2.16 and 1.68 dm^3. Forgetting the oxygen already inside X (burning "C_2H_6") gives 2.52 dm^3; using 5 mol of O_2 per mole of X gives 3.60 dm^3.

Question 10. Answer **E**.

The graphs show only the first few ionisation energies, so they need not show every electron. The big jumps give the number of electrons in each outer shell.

For X, the jump comes after the second electron, so X has two outer electrons and is in Group 2. X has at least six electrons, because six ionisation energies exist. The third to sixth electrons come from one inner shell, which therefore holds at least four electrons. That cannot be the first shell (it holds only two), so there is another shell beneath it. Hence X has at least three shells, and X is magnesium $(2, 8, 2)$ or calcium $(2, 8, 8, 2)$, not beryllium. Statement I is true.

For Y, the jump comes after the sixth electron, so Y is in Group 16. Y has at least eight electrons, so Y is oxygen ($2, 6$, where the graph happens to show every electron) or sulfur $(2, 8, 6)$.

Statement II holds for magnesium and sulfur but fails for magnesium and oxygen, so it need not be true. Statement III is true in every case: X forms X^{2+} and Y forms Y^{2-}, giving the ionic compounds MgO, MgS, CaO or CaS. The answer is I and III only. Assuming that X has only six electrons leads to the impossible arrangement of two electrons outside a first shell of four, which is the clue that the graph of X is incomplete.

Question 11. Answer **A**.

The graph shows amounts, not concentrations. Squeezing the container makes every *concentration* jump at the instant of the change, but the number of moles of each gas cannot change until reaction occurs. So a volume change, like a temperature change, shows up on this graph only as a gradual drift to a new equilibrium, with no jump.

The drift here is N_2O_4 up by 0.06 mol and NO_2 down by 0.12 mol, so the position of equilibrium has moved to the left.

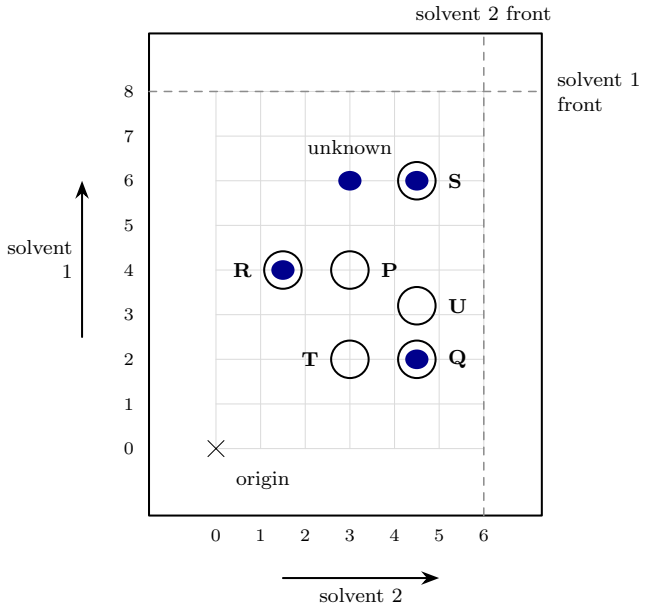
I: a smaller volume raises the pressure, which favours the side with fewer gas molecules: one N_2O_4 against two NO_2, so the left. Possible. II: the forward reaction is endothermic, so the backward reaction is exothermic, and lowering the temperature favours the exothermic direction, the left. Possible. III: a catalyst speeds up the forward and backward reactions equally and does not move the position of equilibrium. Not possible.

Rejecting I because a volume change would make the lines jump is the trap: that is true of a concentration graph only.

Question 12. Answer **E**.

The key is that each solvent has its own front: solvent 1 moved 8.0 cm up the page and solvent 2 moved 6.0 cm across it. So a dye with values $R_f(1)$ and $R_f(2)$ must appear at the point $6.0\,R_f(2)$ cm across and $8.0\,R_f(1)$ cm up. Rather than converting spots into R_f values, plot where each reference dye would be and look for spots there.

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The four spots are at (across, up) $= (4.5, 2.0)$, $(1.5, 4.0)$, $(3.0, 6.0)$ and $(4.5, 6.0)$ cm, so their R_f pairs (solvent 1, solvent 2) are $(0.25, 0.75)$, $(0.50, 0.25)$, $(0.75, 0.50)$ and $(0.75, 0.75)$. These match Q, R and S in both solvents. The spot at $(3.0, 6.0)$ agrees with S in solvent 1 and with P and T in solvent 2, but with no dye in both, so it is a substance that is not among the references. A substance is identified only if it matches in both solvents.

So Q, R and S are present, and there are at least 4 substances (one spot could hide two substances that happen to move identically). Solvent 1 alone could not separate the two spots that are 6.0 cm up, which is the reason for using a second solvent; reading only solvent 2 wrongly matches the unknown spot to P or T.

Question 13. Answer **B**.

Equal volumes of gases contain equal numbers of molecules, so the volumes can be used directly as amounts. Per 20 cm^3 of X:

water $= 115 - 65 = 50$ cm^3; carbon dioxide $= 65 - 25 = 40$ cm^3; the final 10 cm^3 is nitrogen; so the oxygen left over is $25 - 10 = 15$ cm^3 and the oxygen used is $60 - 15 = 45$ cm^3.

One molecule of X therefore gives 2 molecules of CO_2, 2.5 of H_2O and 0.5 of N_2, so X contains 2 C, 5 H and 1 N. Oxygen comes by conservation of atoms, since the oxygen in X cannot be measured directly: per 20 units of X, oxygen atoms in the products are $2(40) + 50 = 130$, of which $2(45) = 90$ came from the oxygen gas used. That leaves 40 oxygen atoms in 20 molecules of X, so 2 per molecule: $C_2H_5NO_2$.

Taking all 25 cm^3 after the sodium hydroxide as oxygen gives $C_2H_5NO_3$; assuming all 60 cm^3 of oxygen was used gives $C_4H_{10}N_2O$; forgetting that H_2O and N_2 each carry two atoms gives $C_4H_5N_2O_4$ or $C_4H_{10}NO_4$; and ignoring the oxygen in X gives C_2H_5N.

Question 14. Answer **E**.

First decide which reagent ran out in each experiment, by comparing them. The same acid gave more gas in Experiment 2, so in Experiment 1 the acid was not used up: there the limestone was limiting, and all of its calcium carbonate reacted. 360 cm^3 is 0.0150 mol, so 2.00 g contains 1.50 g of $CaCO_3$: the limestone is 75% calcium carbonate.

If the limestone had also been limiting in Experiment 2, doubling it would have doubled the gas to 720 cm^3. Only 480 cm^3 (0.0200 mol) was formed, so there the acid ran out: 0.0400 mol of HCl in 50.0 cm^3, which is 0.800 mol dm^{-3}.

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Now test both limits for the new mixture. 3.00 g of limestone contains 2.25 g = 0.0225 mol of $CaCO_3$, enough for 540 cm^3. 40.0 cm^3 of acid contains 0.0320 mol of HCl, enough for 0.0160 mol = 384 cm^3. The smaller wins: 384 cm^3.

540 cm^3 forgets to check the acid. Working out the purity from Experiment 2 (where the limestone was in excess) gives 50% and then 360 cm^3; working out the acid concentration from Experiment 1 gives 0.600 mol dm^{-3} and then 288 cm^3. 768 cm^3 ignores the 1 : 2 ratio.

Question 15. Answer **A**.

The tests identify water (I), Fe^{2+} (the green precipitate; the orange-brown surface is iron(II) hydroxide being oxidised by air to iron(III) hydroxide, which happens only after the test), NH_4^+ (III), SO_4^{2-} (IV) and no halide (V).

Each portion contains 0.784 g of X. The precipitate is $BaSO_4$ ($M_r = 233$): $0.932/233 = 0.004$ mol of SO_4^{2-}. Ammonia: $96/24\,000 = 0.004$ mol of NH_4^+.

Iron was never measured, and it does not need to be: the salt is neutral, so the positive and negative charges balance. $0.004 \times 1 + 2\,n(Fe^{2+}) = 0.004 \times 2$, so $n(Fe^{2+}) = 0.002$ mol.

Mass of ions: $0.004 \times 18 + 0.002 \times 56 + 0.004 \times 96 = 0.072 + 0.112 + 0.384 = 0.568$ g. The water is $0.784 - 0.568 = 0.216$ g = 0.012 mol. The ratio $NH_4^+ : Fe^{2+} : SO_4^{2-} : H_2O = 2 : 1 : 2 : 6$, so X is $(NH_4)_2Fe(SO_4)_2{\cdot}6H_2O$, with 6 water molecules per iron ion.

Taking the iron as Fe^{3+} because of the brown colour gives 0.00133 mol of iron and about 10.6 water molecules per iron ion, which is not a whole number (so 11 is a guess, not an answer). Leaving the ammonium ions out of the mass gives 8; leaving out the iron, which was never weighed, gives about 9; counting the sulfate as sulfur alone gives about 13; dividing the water by the sulfate or ammonium amount gives 3.

Question 16. Answer **D**.

Use scaling rather than finding ΔH or the acid concentration. The temperature rise is (energy released)/(mass $\times\, c$), and the energy released is proportional to the moles of water formed. So the rise at neutralisation is proportional to (moles of water)/(total mass of mixture).

First run (from M): 25.0 cm^3 of 2.00 mol dm^{-3} NaOH is 0.0500 mol, so 0.0500 mol of water forms in 50.0 g of mixture, and the rise is 13.5 K.

Repeat: twice as much acid needs twice as much NaOH, 0.100 mol, which at 4.00 mol dm^{-3} is again 25.0 cm^3. Now 0.100 mol of water forms in 75.0 g. The rise is $13.5 \times 2 \times \frac{50}{75} = 18.0$ K, so the highest temperature is 38.0 °C. The fact that sulfuric acid is diprotic never matters: it is the same acid both times.

47.0 °C doubles the rise but forgets the extra mass; 33.5 °C assumes 50.0 cm^3 of the stronger alkali is needed (mass 100 g); 29.0 °C keeps the moles of water at 0.0500; 36.5 °C scales the highest measured temperature, 32.4 °C, instead of M; 30.8 °C forgets that the doubled acid needs double the alkali, using 12.5 cm^3 of NaOH, 0.0500 mol of water and 62.5 g.

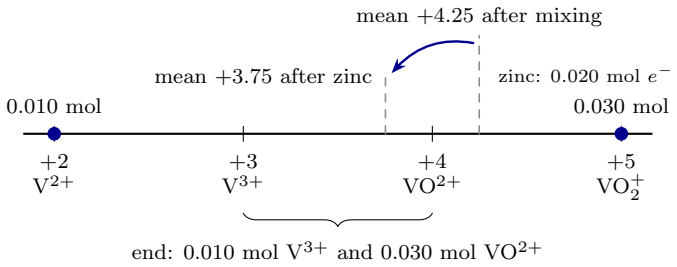
Question 17. Answer **B**.

Do not follow the reactions one at a time. Electrons are only passed around, so the total of the vanadium oxidation states changes only by the electrons the zinc supplies; the rule about adjacent oxidation states then fixes the final mixture from its average.

Oxidation states (not charges): in NH_4VO_3 the ammonium ion is +1, so $V + 3(-2) = -1$ gives +5. In VO_2^+, $V + 2(-2) = +1$ gives +5; in VO^{2+}, $V - 2 = +2$ gives +4. Amount of NH_4VO_3 ($M_r = 117$): $3.51/117 = 0.030$ mol.

Total oxidation number at the start: $0.030 \times 5 + 0.010 \times 2 = 0.170$ mol. Zinc: $0.65/65 = 0.010$ mol, each atom giving 2 electrons, so 0.020 mol of electrons, which lowers the total to 0.150 mol. Shared among 0.040 mol of vanadium, the average oxidation state is +3.75.

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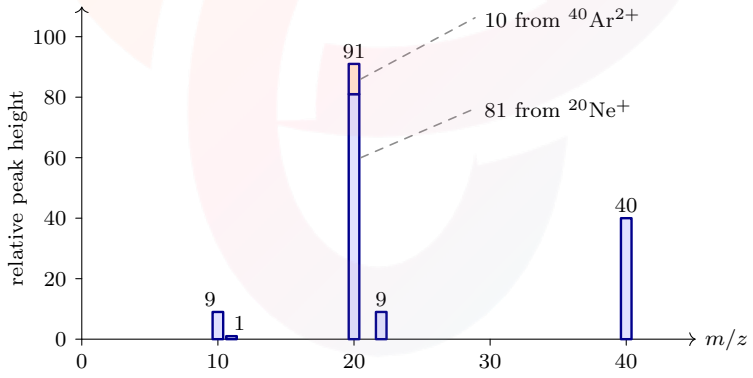
The only allowed final state with average +3.75 is a mixture of +4 and +3: with x mol of VO^{2+} and y mol of V^{3+}, $x + y = 0.040$ and $4x + 3y = 0.150$, so $x = 0.030$ and $y = 0.010$. The order in which the reactions happen makes no difference.

Ignoring the zinc gives the mixture with average +4.25 (VO_2^+ and VO^{2+}); counting only one electron per zinc atom gives average +4.00 (VO^{2+} only); forgetting that V^{2+} and the higher states react with each other leaves three ions, which the stated rule forbids.

Question 18. Answer **A**.

First assign every peak. The peak at $m/z = 40$ is $^{40}Ar^+$, the peak at 22 is $^{22}Ne^+$, and the peaks at 10 and 11 are $^{20}Ne^{2+}$ and $^{22}Ne^{2+}$. The peak at 20 is the awkward one: it contains $^{20}Ne^+$ and also $^{40}Ar^{2+}$, which has the same mass-to-charge ratio.

The way in is to calibrate with the isotope that has no interference. For ^{22}Ne, the 1+ and 2+ peaks are 9 and 1, so neon splits 9 : 1 between 1+ and 2+ ions. The same split applies to ^{20}Ne, whose 2+ peak is 9, so its 1+ ions contribute 81 to the peak at 20. The remaining $91 - 81 = 10$ is $^{40}Ar^{2+}$.



Neon: $9 + 1 + 81 + 9 = 100$. Argon: $40 + 10 = 50$. The ratio is $100 : 50 = 2 : 1$. (As a check, neon is 90% ^{20}Ne, close to its natural composition.)

Giving the whole of the peak at 20 to neon gives $110 : 40 = 11 : 4$. Ignoring all the 2+ ions gives $100 : 40 = 5 : 2$ (all of the peak at 20 treated as neon) or $90 : 40 = 9 : 4$ (correct split, but only 1+ ions counted). Counting argon's 2+ ions but not neon's gives $90 : 50 = 9 : 5$.

Question 19. Answer **C**.

Write the hydride as XH_n, with $n = 1$ or 2. The reaction is $XH_n + nH_2O \rightarrow X(OH)_n + nH_2$; in ionic terms each hydride ion takes a hydrogen ion from a water molecule, $H^- + H_2O \rightarrow OH^- + H_2$. So one hydride ion gives one hydrogen molecule, whatever n is.

Hydrogen produced: $720/24\,000 = 0.030$ mol, so the sample contains 0.030 mol of H^-. The mass of hydride per mole of H^- is $0.63/0.030 = 21$ g, of which 1 g is hydrogen, so X contributes 20 g for every hydride ion: $A_r(X) = 20n$. With $n = 1$ there is no Group 1 metal of relative atomic mass 20; with $n = 2$, $A_r = 40$, which is calcium. Statement I is true, and testing the six possible hydrides one by one is unnecessary.

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Statement II: H^- has one proton and two electrons, configuration 2, the same as helium. True.

Statement III: each H_2 molecule is made from one hydride ion and one hydrogen atom taken from water. True.

Statement IV: the hydride ion is the negative ion, so it moves to the positive electrode (anode) and is oxidised there, $2H^- \rightarrow H_2 + 2e^-$, while calcium forms at the negative electrode. Hydrogen appears at the anode, so IV is false. The answer is I, II and III only.

Question 20. Answer **C**.

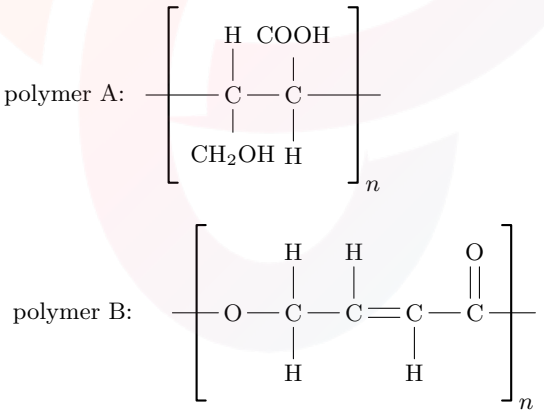
Two observations make this short. At the methyl orange end point the solution is just sodium chloride, so the total acid used, $22.5 + 2.5 = 25.0$ cm^3, contains one H^+ for every sodium ion, that is, for every NaOH originally present. And the second stage converts only hydrogencarbonate, and every hydrogencarbonate came from one carbonate, so the second titre, 2.5 cm^3, measures the carbonate.

Each carbonate ion used up two NaOH, so the NaOH that reacted corresponds to $2 \times 2.5 = 5.0$ cm^3 of acid, out of 25.0 cm^3 for all the NaOH originally present: $5.0/25.0 = 20\%$. In moles: carbonate $= 2.5 \times 10^{-4}$, hydroxide left $= (22.5 - 2.5) \times 10^{-4} = 2.0 \times 10^{-3}$, original NaOH $= 2.0 \times 10^{-3} + 2 \times 2.5 \times 10^{-4} = 2.5 \times 10^{-3}$ mol. No concentration of the original solution is needed, because the answer is a ratio.

Forgetting that each carbonate used two NaOH gives 10%; comparing with the first titre only gives 11.1% or 22.2%; forgetting that the first titre also includes one H^+ per carbonate gives 12.5% or 25%; 80% is the fraction that has not reacted.

Question 21. Answer **F**.

First draw both repeat units. M is $HOCH_2CH{=}CHCOOH$, $C_4H_6O_3$, $M_r = 102$.



In A only the double bond opens, so the repeat unit $-CH(CH_2OH)-CH(COOH)-$ has the same formula as M, $C_4H_6O_3$ (102), and keeps both the OH and COOH groups. In B each monomer loses one H_2O into an ester link, so the repeat unit $-O-CH_2-CH{=}CH-CO-$ is $C_4H_4O_2$ (84), with the double bond kept.

I: 1.02 g of A is 0.0100 mol of repeat units. Sodium replaces the H of every OH and every COOH group (the acid group reacts with the metal too), and every two such H atoms give one H_2. So 0.0200 mol of reactive H gives 0.0100 mol of H_2, which is 240 cm^3. True. (Counting only one of the two groups gives 120 cm^3.)

II: B still has C=C bonds in its chain; A has none. True.

III: hydrolysis breaks the ester links of B and gives back M, but the chain of A is held together by carbon to carbon bonds, which hydrolysis does not break. False.

IV: both repeat units contain 4 carbon atoms, but in different masses. Per gram, B contains $\frac{102}{84}$ times as much carbon as A, so it gives more CO_2. False.

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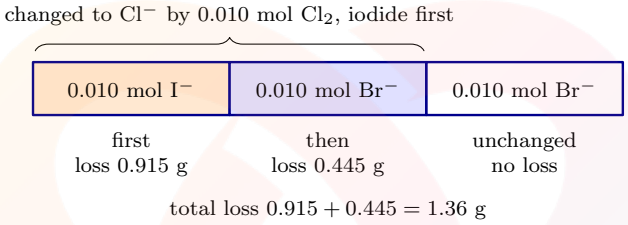
V: 10.2 g of M is 0.100 mol; for very long chains one water molecule is lost per monomer, so 0.100 mol of water, 1.8 g, is formed. True.

The answer is I, II and V only.

Question 22. Answer **E**.

The chlorine is $240/24\,000 = 0.010$ mol of Cl_2, so it turns exactly 0.020 mol of halide ions into chloride ions. The question is which halide ions. Bromine is more reactive than iodine, so any bromine formed while iodide ions remain would at once oxidise them: $Br_2 + 2I^- \rightarrow 2Br^- + I_2$. So in the end the iodide ions are used up before any bromide ions are left oxidised.

Now track mass, not moles of each salt. Replacing I^- by Cl^- loses $127 - 35.5 = 91.5$ g mol^{-1}; replacing Br^- by Cl^- loses 44.5 g mol^{-1}. If all 0.020 mol of the swaps were iodide, the loss would be 1.83 g. The actual loss, $4.04 - 2.68 = 1.36$ g, is smaller, so all the iodide (say n mol) was used up and the rest of the swaps were bromide. Count every swap at 44.5 and add 47 extra for each iodide: $1.36 = 0.020 \times 44.5 + 47n = 0.89 + 47n$, so $n = 0.010$ mol, which is $0.010 \times 166 = 1.66$ g of potassium iodide.



The 4.04 g is needed only as a check: it leaves 2.38 g (0.020 mol) of potassium bromide, enough to supply the 0.010 mol of bromide that reacted.

Putting the whole loss down to iodide gives $1.36/91.5 \times 166 \approx 2.47$ g. Treating the chlorine as 0.010 mol of single atoms gives 3.23 g, which is more iodide than that chlorine could have reacted with. Assuming the chlorine shares itself between bromide and iodide in proportion to their amounts gives 2.35 g. The values 2.38 g and 1.27 g are the mass of potassium bromide and the mass of iodine in the potassium iodide.

Question 23. Answer **F**.

Keep two accounts, copper(II) sulfate and water, and remember that water of crystallisation moves with the crystals. $CuSO_4{\cdot}5H_2O$ has $M_r = 250$, so 100 g of it is 64 g of $CuSO_4$ and 36 g of water. The solution therefore holds 64 g of $CuSO_4$ in $142 + 36 = 178$ g of water, which is about 36 g per 100 g of water: well below the solubility at 80 °C (about 55 g), so it all dissolves.

At 20 °C the graph gives 20 g per 100 g of water, so the saturated solution has $CuSO_4$: water = 1 : 5 by mass. The crystals have $CuSO_4$: water = 160 : 90 = 16 : 9. Write the crystals as $16k$ g + $9k$ g and the solution as s g + $5s$ g:

$16k + s = 64, \qquad 9k + 5s = 178.$

Five times the first minus the second gives $71k = 142$, so $k = 2$ and the crystals weigh $25k = 50.0$ g. Check: 50 g of crystals hold 32 g of $CuSO_4$ and 18 g of water, leaving 32 g in 160 g of water, which is 20 g per 100 g.

44.4 g comes from finding the 28.4 g of $CuSO_4$ that must leave the solution if the water stayed at 178 g and converting it to crystals, forgetting that the crystals also take 90/160 of that mass in water out of the solution; 28.4 g stops before converting. Ignoring water of crystallisation entirely gives 71.6 g; forgetting the 36 g of water released from the dissolved crystals gives 55.6 g or 62.7 g.

Question 24. Answer **D**.

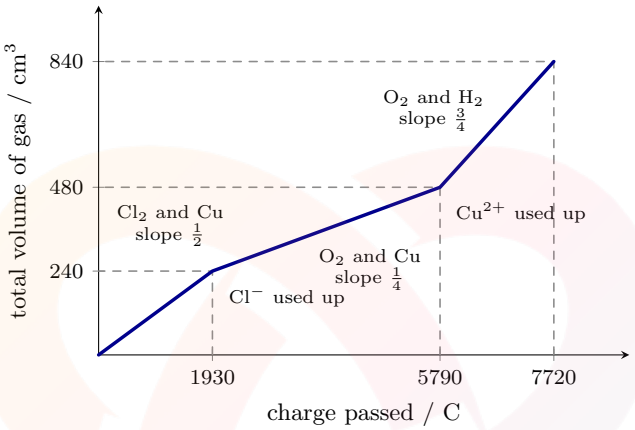
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Read the graph through its gradient: moles of gas per mole of electrons. Each possible pair of electrode products has its own value.

Anode: $2Cl^- \to Cl_2 + 2e^-$ gives $\frac{1}{2}$ mol of gas per mole of electrons; after the chloride is used up, $4OH^- \to O_2 + 2H_2O + 4e^-$ gives $\frac{1}{4}$. Cathode: $Cu^{2+} + 2e^- \to Cu$ gives no gas; after the copper ions are used up, hydrogen forms, $\frac{1}{2}$ mol per mole of electrons. Sodium ions are never discharged.

At first the products are chlorine and copper: gradient $\frac{1}{2}$. If the copper ions ran out first, hydrogen would join the chlorine and the gradient would rise to 1. If the chloride ran out first, oxygen would replace the chlorine and the gradient would fall to $\frac{1}{4}$. Finally, oxygen and hydrogen give $\frac{1}{4} + \frac{1}{2} = \frac{3}{4}$.

Charges in moles of electrons: 1930 C = 0.020 mol, 5790 C = 0.060 mol, 7720 C = 0.080 mol. The gradients are $\frac{0.010}{0.020} = \frac{1}{2}$, then $\frac{0.010}{0.040} = \frac{1}{4}$, then $\frac{0.015}{0.020} = \frac{3}{4}$ (gas volumes 240, 240 and 360 cm^3, each divided by 24 000). The fall to $\frac{1}{4}$ shows that the chloride ran out first.



So the chloride was used up after 0.020 mol of electrons: 0.020 mol of NaCl, $0.020 \times 58.5 = 1.17$ g. The copper ions were used up after 0.060 mol of electrons in total, so there were $0.060/2 = 0.030$ mol of Cu^{2+}: $0.030 \times 160 = 4.80$ g of $CuSO_4$.

Assuming the first kink is where the copper runs out gives 3.51 g and 1.60 g, but then the gradient would have risen. Using one electron per copper ion gives 9.60 g; counting only the copper deposited during the oxygen stage gives 3.20 g; reading the final point as the end of the copper gives 6.40 g; finding the chloride from the 240 cm^3 of Cl_2 without doubling gives 0.585 g.

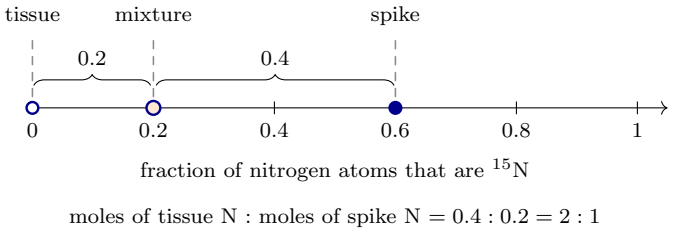
Question 25. Answer **C**.

The spike. Its nitrogen has relative atomic mass $0.6 \times 15 + 0.4 \times 14 = 14.6$, so M_r of the spike is $2(14.6 + 4) + 32 + 4 \times 16 = 133.2$. The spike is $1.332/133.2 = 0.0100$ mmol, containing 0.0200 mmol of nitrogen atoms, of which 0.0120 mmol are ^{15}N.

The spectrum. Count atoms, not molecules. Out of every 25 molecules (50 atoms), the 8 at $m/z = 29$ each contain one ^{15}N and the 1 at $m/z = 30$ contains two, so 10 atoms in 50, a fraction $\frac{1}{5}$, are ^{15}N. (Random pairing with $\frac{1}{5}$ gives 16 : 8 : 1, as shown.)

The balance. Every ^{15}N atom came from the spike, so the mixture contains $0.0120 \times 5 = 0.0600$ mmol of nitrogen. Subtracting the spike's 0.0200 mmol leaves 0.0400 mmol from the tissue, which is $0.0400 \times 14 = 0.56$ mg. Equivalently, the mixture's ^{15}N fraction is a weighted average of 0 (tissue) and 0.6 (spike), and the lever below gives tissue nitrogen : spike nitrogen = 2 : 1.

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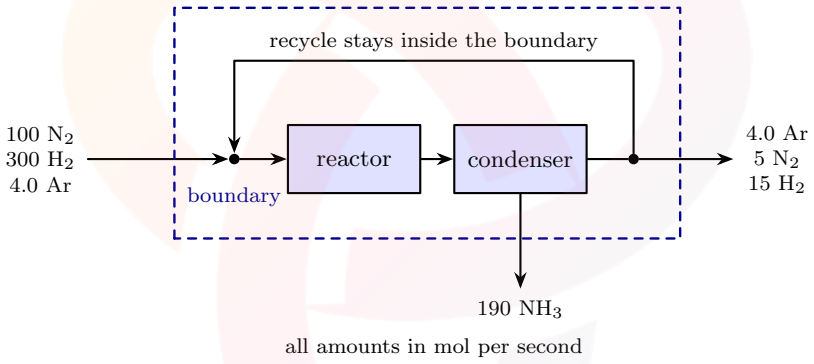


moles of tissue N : moles of spike N $= 0.4 : 0.2 = 2 : 1$

Treating the spike as pure ^{15}N gives 1.12 mg. Subtracting only the spike's ^{15}N instead of all its nitrogen gives $0.048 \times 14 \approx 0.67$ mg, and forgetting to subtract the spike at all gives 0.84 mg. Using one nitrogen atom per formula unit of ammonium sulfate halves the answer to 0.28 mg. Taking $\frac{9}{25}$, the fraction of molecules containing some ^{15}N, as the atom fraction gives about 0.19 mg, and taking $\frac{1}{25}$ from the peak at 30 alone gives 3.92 mg.

Question 26. Answer **A**.

Do not follow the gas round the loop. Draw a boundary round the whole plant: at steady state, whatever enters it must leave it. The recycle stream never crosses the boundary, so the 15% per pass is irrelevant.



Argon is the key, because it enters only in the feed and leaves only in the purge. So the purge carries 4.0 mol of argon each second, and with it $5 \times 4.0 = 20$ mol of nitrogen and hydrogen. Nitrogen and hydrogen enter in the ratio $1 : 3$ and react in the ratio $1 : 3$, so they are in the ratio $1 : 3$ everywhere, and the purge loses 5 mol of N_2 and 15 mol of H_2.

The rest reacts: 95 mol of N_2 and 285 mol of H_2 each second, giving $2 \times 95 = 190$ mol of NH_3, which is $190 \times 17 = 3230$ g $= 3.23$ kg.

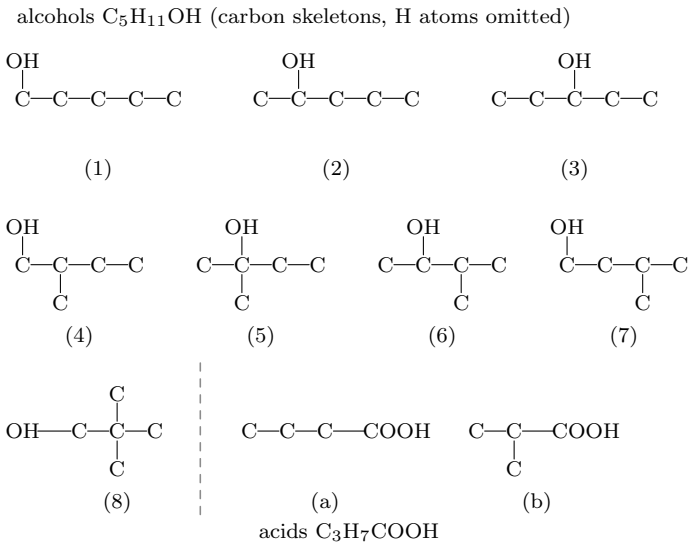
Reading the purge as 20% argon gives 3.26 kg; ignoring the purge gives 3.40 kg; applying 15% to the fresh feed once gives 0.51 kg; taking the moles of ammonia equal to the 380 mol of nitrogen and hydrogen reacted gives 6.46 kg, and equal to the 95 mol of nitrogen reacted gives 1.62 kg.

Question 27. Answer **A**.

An acid with n carbon atoms is $C_nH_{2n}O_2$ and an alcohol with m carbon atoms is $C_mH_{2m+2}O$. The neat observation is that CH_4 and O have the same mass, 16: the alcohol $C_{n+1}H_{2n+4}O$ is the acid $C_nH_{2n}O_2$ with one O swapped for CH_4. So equal masses means that the alcohol has one more carbon atom than the acid. (Algebraically, $14n + 32 = 14m + 18$ gives $m = n + 1$.)

Ester formation loses one water, so $158 = 2M - 18$, where M is the common relative formula mass, and $M = 88$. Then $14n + 32 = 88$ gives $n = 4$ and $m = 5$: the acid is C_3H_7COOH and the alcohol is $C_5H_{11}OH$.

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Acids: the COOH carbon is at the end of the chain, and the remaining three carbons are either a straight chain or branched: butanoic acid and 2-methylpropanoic acid, 2 acids.

Alcohols, sorted by the longest chain: five-carbon chain, OH on carbon 1, 2 or 3 (pentan-1-ol, pentan-2-ol, pentan-3-ol); four-carbon chain with a methyl branch, OH at each distinct position (2-methylbutan-1-ol, 2-methylbutan-2-ol, 3-methylbutan-2-ol, 3-methylbutan-1-ol); three-carbon chain with two methyl branches, 2,2-dimethylpropan-1-ol. That is 8 alcohols, so E has $2 \times 8 = 16$ possible structures.

Reversing the conclusion (acid with five carbons, alcohol with four) gives 4 and 4, but then the masses would be 102 and 74, not equal. Missing 2,2-dimethylpropan-1-ol gives 7; counting only unbranched alcohols gives 3.

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